

Computer Forensics



Unit – 3

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Unit – 3

Windows Forensics



File Allocation Table



FAT: Introduction



- The File Allocation Table (FAT) file system is one of the most simple file systems found in common operating systems.
- FAT is the primary file system of the Microsoft DOS and Windows 9x operating systems, but the NT, 2000, and XP line has defaulted to the New Technologies File System (NTFS).
- FAT is supported by all Windows and most Unix operating systems.



FAT: Introduction



- FAT is frequently found in compact flash cards for digital cameras and USB “thumb drives.”
- The basic concept of a FAT file system is that each file and directory is allocated a data structure, called a directory entry, that contains the file name, size, starting address of the file content, and other metadata.
- File and directory content is stored in data units called clusters. If a file or directory has allocated more than one cluster, the other clusters are found by using a structure that is called the FAT structure.



FAT: Introduction



- The FAT structure is used to identify the next cluster in a file, and it is also used to identify the allocation status of clusters.
- Therefore it is used in both the content and metadata categories.
- There are three different versions of FAT: FAT₁₂, FAT₁₆, and FAT₃₂.
- The major difference among them is the size of the entries in the FAT structure.



FAT: Introduction



- In a FAT file system, the file system category of data can be found in the boot sector data structure.
- The boot sector is located in the first sector of the volume, and it is part of the reserved area of the file system.
- There is no field that identifies the file system as FAT₁₂, FAT₁₆, or FAT₃₂.
- The type can be determined only by performing calculations from the boot sector data.



FAT: Introduction



- A FAT₃₂ file system boot sector contains additional data, including the sector address of a backup copy of the boot sector and a major and minor version number.
- The backup copy of the boot sector can be used if the version in sector 0 becomes corrupt, and the Microsoft documentation says that it should always be in sector 6 so that tools can automatically find it if the default copy goes bad.



FAT: Introduction



- FAT₃₂ file systems also have an FSINFO data structure that contains information about the location of the next available cluster and the total amount of free clusters.



FAT: Introduction



- The layout of the FAT file system has three physical sections:
 - The first section is the reserved area, and it includes data in the file system category. In FAT12 and FAT16 this area is typically only 1 sector in size, but the size is defined in the boot sector.
 - The second section is the FAT area, and it contains the primary and backup FAT structures. It starts in the sector following the reserved area, and its size is calculated based on the number and size of FAT structures.
 - The third section is the data area, and it contains the clusters that will be allocated to store file and directory content.



FAT: Data structure (36Bytes)



Byte Range	Description	Essential
0-2	Assembly instruction to jump to boot code.	No (unless it is a bootable file system)
3-10	OEM Name in ASCII.	No
11-12	Bytes per sector. Allowed values include 512, 1024, 2048, and 4096.	Yes
13-13	Sectors per cluster (data unit). Allowed values are powers of 2, but the cluster size must be 32KB or smaller.	Yes
14-15	Size in sectors of the reserved area.	Yes
16-16	Number of FATs. Typically two for redundancy, but according to Microsoft it can be one for some small storage devices.	Yes
17-18	Maximum number of files in the root directory for FAT12 and FAT16. This is 0 for FAT32 and typically 512 for FAT16.	Yes



FAT: Data structure (36Bytes)



Byte Range	Description	Essential
19-20	16-bit value of number of sectors in file system. If the number of sectors is larger than can be represented in this 2-byte value, a 4-byte value exists later in the data structure and this should be 0.	Yes
21-21	Media type. According to the Microsoft documentation, 0xf8 should be used for fixed disks and 0xf0 for removable.	No
22-23	16-bit size in sectors of each FAT for FAT12 and FAT16. For FAT32, this field is 0.	Yes
24-25	Sectors per track of storage device.	No
26-27	Number of heads in storage device.	No
28-31	Number of sectors before the start of partition.1	No
32-35	32-bit value of number of sectors in file system. Either this value or the 16-bit value above must be 0.	Yes



FAT12/16: Data structure



Byte Range	Description	Essential
36-36	BIOS INT _{13h} drive number.	No
37-37	Not used.	No
38-38	Extended boot signature to identify if the next three values are valid. The signature is 0x29.	No
39-42	Volume serial number, which some versions of Windows will calculate based on the creation date and time.	No
43-53	Volume label in ASCII. The user chooses this value when creating the file system.	No
54-61	File system type label in ASCII. Standard values include "FAT," "FAT12," and "FAT16," but nothing is required.	No
62-509	Not used.	No
510-511	Signature value (0xAA55).	No



FAT32: Data structure



Byte Range	Description	Essential
36-39	32-bit size in sectors of one FAT.	Yes
40-41	Defines how multiple FAT structures are written to. If bit 7 is 1, only one of the FAT structures is active and its index is described in bits 0-3. Otherwise, all FAT structures are mirrors of each other.	Yes
42-43	The major and minor version number	Yes
44-47	Cluster where root directory can be found.	Yes
48-49	Sector where FSINFO structure can be found.	No
50-51	Sector where backup copy of boot sector is located (default is 6).	No
52-63	Reserved.	No
64-64	BIOS INT _{13h} drive number.	No



FAT32: Data structure



Byte Range	Description	Essential
65-65	Not used.	No
66-66	Extended boot signature to identify if the next three values are valid. The signature is 0x29.	No
67-70	Volume serial number, which some versions of Windows will calculate based on the creation date and time.	No
71-81	Volume label in ASCII. The user chooses this value when creating the file system.	No
82-89	File system type label in ASCII. Standard values include "FAT32," but nothing is required.	No
90-509	Not used.	No
510-511	Signature value (0xAA55).	No



FAT: Data structure



Byte Range	Description	Essential
0-0	First character of file name in ASCII and allocation status (0xe5 or 0x00 if unallocated)	Yes
1-10	Characters 2 to 11 of file name in ASCII.	Yes
11-11	File Attributes	Yes
12-12	Reserved	No
13-13	Created time (tenths of second)	No
14-15	Created time (hours, minutes, seconds)	No
16-17	Created day	No



FAT: Data structure



Byte Range	Description	Essential
18-19	Accessed day	No
20-21	High 2 bytes of first cluster address (0 for FAT12 and FAT16)	Yes
22-23	Written time (hours, minutes, seconds)	No
24-25	Written day	No
26-27	Low 2 bytes of first cluster address	Yes
28-31	Size of file (0 for directories)	Yes



FAT: Data structure



Flag values for the directory entry attributes field.		
Flag Value (in bits)	Description	Essential
0000 0001 (0x01)	Read only	No
0000 0010 (0x02)	Hidden file	No
0000 0100 (0x04)	System file	No
0000 1000 (0x08)	Volume label	Yes
0000 1111 (0x0f)	Long file name	Yes
0001 0000 (0x10)	Directory	Yes
0010 0000 (0x20)	Archive	No



FAT: Data structure



Data structure for an LFN FAT directory entry.		
Byte Range	Description	Essential
0-0	Sequence number (ORed with 0x40) and allocation status (0xe5 if unallocated)	Yes
1-10	File name characters 1-5 (Unicode)	Yes
11-11	File attributes (0x0f)	Yes
12-12	Reserved	No
13-13	Checksum	Yes
14-25	File name characters 6-11 (Unicode)	Yes
26-27	Reserved	No
28-31	File name characters 12-13 (Unicode)	Yes



ExFAT: Data structure (36Bytes)



Byte Range	Description	Essential
0-2	jump instruction for CPUs	No (unless it is a bootable file system)
3-10	name of the file system on the volume. "EXFAT "	No
11-63	range of bytes the packed BIOS parameter block consumes on FAT12/16/32 volumes. Must be 0.	Yes
72-79	media-relative sector offset of the partition which hosts the given exFAT volume.	Yes
80-83	size of the given exFAT volume in sectors.	Yes
84-87	sector offset of the First FAT	Yes
88-91	length, in sectors, of each FAT table	Yes



ExFAT: Data structure (36Bytes)



Byte Range	Description	Essential
92-95	sector offset of the Cluster Heap	No (unless it is a bootable file system)
96-99	number of clusters the Cluster Heap contains.	No
100-103	cluster index of the first cluster of the root directory.	Yes
104-105	unique serial number	Yes
106-107	major and minor revision numbers of the exFAT structures on the given volume.	Yes
108-108	flags which indicate the status of various file system structures on the ExFAT volume.	Yes
109-109	describe which FAT and Allocation Bitmap are active	Yes



ExFAT: Data structure (36Bytes)



Byte Range	Description	Essential
110-110	describe whether the volume is dirty or not. 0, which means the volume is probably in a consistent state 1, which means the volume is probably in an inconsistent state	No (unless it is a bootable file system)
111-111	describe whether an implementation has discovered media failures or not 0, which means the hosting media has not reported failures or any known failures are already recorded in the FAT as "bad" clusters 1, which means the hosting media has reported failures (i.e. has failed read or write operations)	No
112-112	no significant meaning	Yes
113-119	bytes per sector expressed as $\log_2(N)$, where N is the number of bytes per sector.	Yes
120-509	sectors per cluster expressed as $\log_2(N)$, where N is number of sectors per cluster.	Yes
510-511	number of FATs and Allocation Bitmaps the volume contains.	Yes
512-512	extended INT 13h drive number	Yes



FAT



- The data between bytes 62 to 509 in a FAT12/16 file system, and bytes 90 to 509 in a FAT32 file system do not have a specified purpose, but are typically used to store boot code and error messages.
- The data is written in little endian ordering, so the data structure fields that are numbers will appear in reverse order and strings will appear in the expected order.



FAT: Structure

```

00000000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06  EX-MSDOS5.0...@
00000001 02 00 00 00 00 F8 00 00-3F 00 FF 00 80 00 00 00  ....e...y....
00000002 80 7F E7 01 E0 3C 00 00-00 00 00 00 02 00 00 00  ..c-à<.....
00000003 01 00 06 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000004 80 00 29 23 61 FB BE 4E-4F 20 4E 41 4D 45 20 20  ..) #aû?NO NAME
00000005 20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4  FAT32 3É·Nwô
00000006 7B 8E C1 8E D9 BD 00 7C-88 56 40 88 4E 02 8A 56  {·À·Ü%·)·V@·N·V
00000007 40 B4 41 BB AA 55 CD 13-72 10 81 FB 55 AA 75 0A  @'A»*UÍ·r·ûU+u·
00000008 F6 C1 01 74 05 FE 46 02-EB 2D 8A 56 40 B4 08 CD  ôÁ·t·pF·ë-·V@·í
00000009 13 73 05 B9 FF FF 8A F1-66 0F B6 C6 40 66 0F B6  ·s·ÿÿ·ñf·EE@f·q
0000000a D1 80 E2 3F F7 E2 86 CD-C0 ED 06 41 66 0F B7 C9  N·â?+â·îÂi·Af·É
0000000b 66 F7 E1 66 89 46 F8 83-7E 16 00 75 39 83 7E 2A  f+áf·Fø~·u9~*
0000000c 00 77 33 66 8B 46 1C 66-83 C0 0C BB 00 80 B9 01  ·w3f·F·f·À·~·¹·
0000000d 00 E8 2C 00 E9 A8 03 A1-F8 7D 80 C4 7C 8B F0 AC  ·è,·é"·;ø}·Äl·ä·
0000000e 84 C0 74 17 3C FF 74 09-B4 0E BB 07 00 CD 10 EB  ·Àt·<ytc·"·»·í·ë
0000000f EE A1 FA 7D EB E4 A1 7D-80 EB DF 98 CD 16 CD 19  i;ú)ëä;·)·ëâ·í·í·
00000010 66 60 80 7E 02 00 0F 84-20 00 66 6A 00 66 50 06  f·~·~·~·fj·fP·
00000011 53 66 68 10 00 01 00 B4-42 8A 56 40 8B F4 CD 13  Sfh·~·~·B·V@·ôí·
00000012 66 58 66 58 66 58 66 58-EB 33 66 3B 46 F8 72 03  fXfXfXfXfXfXf;Fer·
00000013 F9 EB 2A 66 33 D2 66 0F-B7 4E 18 66 F7 F1 FE C2  ûë*f3ôf·N·f+ñpÂ
00000014 8A CA 66 8B D0 66 C1 EA-10 F7 76 1A 86 D6 8A 56  ·È·BfÂë·+v··ô·V
00000015 40 8A E8 C0 E4 06 0A CC-B8 01 02 CD 13 66 61 0F  @·ëÄä··î··í·fa·
00000016 82 74 FF 81 C3 00 02 66-40 49 75 94 C3 42 4F 4F  ·ty·Ä·f@Iu·ÄBOO
00000017 54 4D 47 52 20 20 20 20-00 00 00 00 00 00 00 00  TMGR .....
00000018 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000019 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
0000001a 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....Di
0000001b 73 6B 20 65 72 72 6F 72-FF 0D 0A 50 72 65 73 73  sk errorÿ·Pres·
0000001c 20 61 6E 79 20 6B 65 79-20 74 6F 20 72 65 73 74  any key to test
0000001d 61 72 74 0D 0A 00 00 00-00 00 00 00 00 00 00 00  art.....
0000001e 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
0000001f 00 00 00 00 00 00 00 00-AC 01 B9 01 00 00 55 AA  .....·¹··U·
00000020 52 52 61 41 00 00 00 00-00 00 00 00 00 00 00 00  RRAA.....
00000021 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000022 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000023 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000024 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000025 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000026 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000027 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
00000028 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00  .....
Cursor pos = 0; log sec = 0; phy sec = 128
    
```

log sec = 0; phy sec = 128

First Sector of Storage Medium

The logo of the National Food Security University (NFSU) is a shield-shaped emblem. At the top, the acronym 'NFSU' is written in a serif font. The shield is divided into four quadrants: the top-left is blue with a white computer monitor icon; the top-right is red with a white fingerprint icon; the bottom-left is red with a white icon of a curved arrow pointing upwards; and the bottom-right is blue with a white DNA double helix icon. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a stylized Devanagari script.



```

61 6E 79 20 6B 65 79-20
72 74 0D 0A 00 00 00-00
log sec = 6; phy sec = 134

```

Backup FAT at 6th Sector of Storage Medium

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a blue background. Inside the shield, there are four white icons: a computer monitor, a fingerprint, a DNA double helix, and a magnifying glass. The text 'NFSU' is written in white at the top of the shield. Below the shield, the motto 'विद्यया अमृतं अश्नुते' is written in Devanagari script.



Assembly instruction to jump to boot code.

FAT: Structure Analysis (3-10)

```

000000000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06 #X MSDOS5.0 ...@
000000010 02 00 00 00 00 00 F8 00 00-3F 00 FF 00 80 00 00 .....ø..?.ÿ.....
000000020 80 7F E7 01 E0 3C 00 00-00 00 00 00 02 00 00 00 ..c.à<.....
000000030 01 00 06 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000040 80 00 29 23 61 FB BE 4E-4F 20 4E 41 4D 45 20 20 ...) #a*NO NAME
000000050 20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4 FAT32 3E-Rwô
000000060 7B 8E C1 8E D9 BD 00 7C-88 56 40 88 4E 02 88 56 ([.A.U4.|-VQ.N-V
000000070 40 B4 41 BB AA 55 CD 13-72 10 81 FB 55 AA 75 50 0'As*U!r.-@U*u.
000000080 F6 C1 01 74 05 FE 46 02-EB 2D 8A 56 40 B4 08 CD A.t.bF.e.-VQ*f
000000090 13 73 05 B9 FF FF 8A F1-66 0F B6 C6 40 66 0F B6 S.YY.Rf.REF.g
0000000a0 D1 80 E2 3F F7 E2 86 CD-C0 ED 06 41 66 0F B7 C9 N.â.â.iAi.Af.-E
0000000b0 66 F7 E1 66 89 46 F8 83-7E 16 00 75 39 83 7E 2A f+âf.f.-u9--*
0000000c0 00 77 33 66 8B 46 1C 66-83 C0 0C BB 00 80 B9 01 w3f.F.f.-a..*
0000000d0 00 E8 2C 00 E9 A8 03 A1-F8 7D 80 C4 7C 8B F0 AC è,è"-;a).-â-
0000000e0 84 C0 74 17 3C FF 74 09-B4 0E BB 07 00 CD 10 EB Àt.<yT.-w.-è
0000000f0 EE A1 FA 7D EB E4 A1 7D-80 EB DF 98 CD 16 CD 19 i;ù)â;}.âb.f.i.
000000100 66 60 80 7E 02 00 0F 84-20 00 66 6A 00 66 50 06 f'-----fj.fP
000000110 53 66 68 10 00 01 00 B4-42 8A 56 40 8B F4 CD 13 Sfh.....B-VQ.ôI.
000000120 66 58 66 58 66 58 66 58-EB 33 66 3B 46 F9 72 03 fXfXfXfXf3f;Fer
000000130 F9 E8 2A 66 33 D2 66 0F-B7 4E 18 66 F7 F1 FE C2 uè*f3ôf.N.f-Rpâ
000000140 8A CA 66 8B D0 66 C1 EA-10 F7 76 1A 86 D6 8A 56 Èf-âfâè.-v.-O.V
000000150 40 8A E8 C0 E4 06 0A CC-B8 01 02 CD 13 66 61 0F 8-ââ.-i.-i.fa
000000160 82 74 FF 81 C3 00 02 66-40 49 75 94 C3 42 4F 4F ty.Â.-f@Iu.âBOO
000000170 54 4D 47 52 20 20 20 20-00 00 00 00 00 00 00 00 TMSR .....
000000180 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000190 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000001a0 00 00 00 00 00 00 00 00-00 00 00 00 0D 0A 44 69 .....Di
0000001b0 73 6B 20 65 72 72 6F 72-FF 0D 0A 50 72 65 73 73 sk error;-Press
0000001c0 20 61 6E 79 20 6B 65 79-20 74 6F 72 65 73 74 any key to rest
0000001d0 61 72 74 0D 0A 00 00 00-00 00 00 00 00 00 00 00 art.....
0000001e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000001f0 00 00 00 00 00 00 00 00-AC 01 B9 01 00 00 55 AA .....-.*U*
    
```

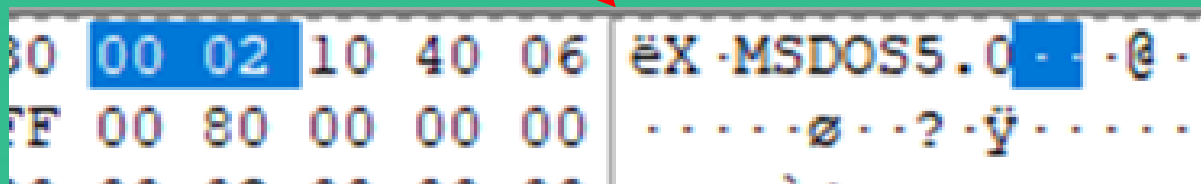
```

000000000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06 #X MSDOS5.0 ...@
000000010 02 00 00 00 00 00 F8 00 00-3F 00 FF 00 80 00 00 00 .....ø..?.ÿ.....
000000020 80 7F E7 01 E0 3C 00 00-00 00 00 00 02 00 00 00 ..c.à<.....
    
```

OEM Name in ASCII.

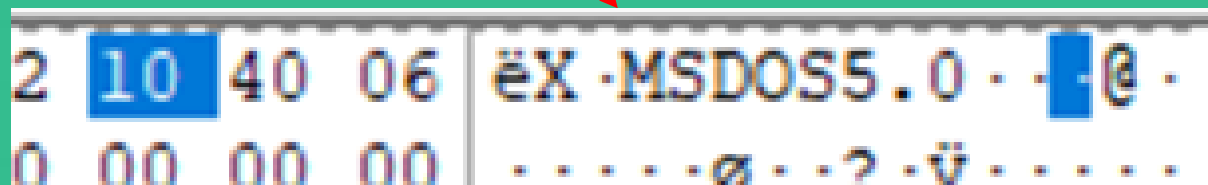


The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

$$= 1 * 512 = 512$$


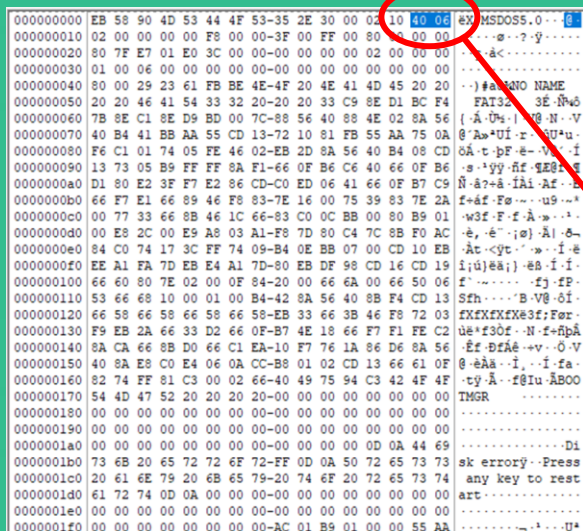
Allowed values include 512, 1024, 2048, and 4096.

The logo of Noida Feature Science University (NFSU) is a shield-shaped emblem. At the top, the letters "NFSU" are written in a bold, sans-serif font. The shield is divided into four quadrants: the top-left quadrant is blue with a white computer monitor icon; the top-right quadrant is red with a white fingerprint icon; the bottom-left quadrant is blue with a white DNA double helix icon; and the bottom-right quadrant is red with a white DNA double helix icon. Below the shield, the Sanskrit motto "विद्यया अमृतं अश्नुते" (Vidyayā amṛtaṁ aśnute) is written in a stylized, flowing font.

$$= 1 * 16 = 16 \text{ Sectors} = 16 * 512 = 8192 = 4\text{KB}$$


Allowed values are powers of 2, but the cluster size must be 32KB or smaller.

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

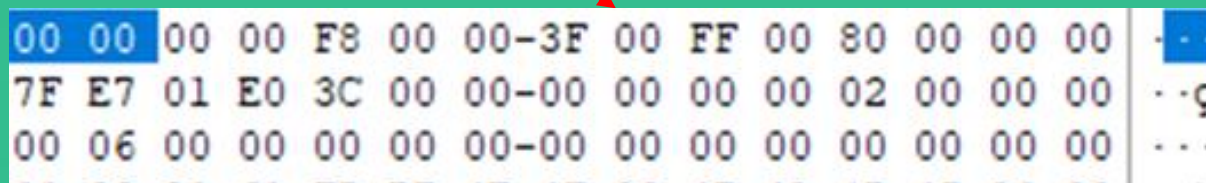
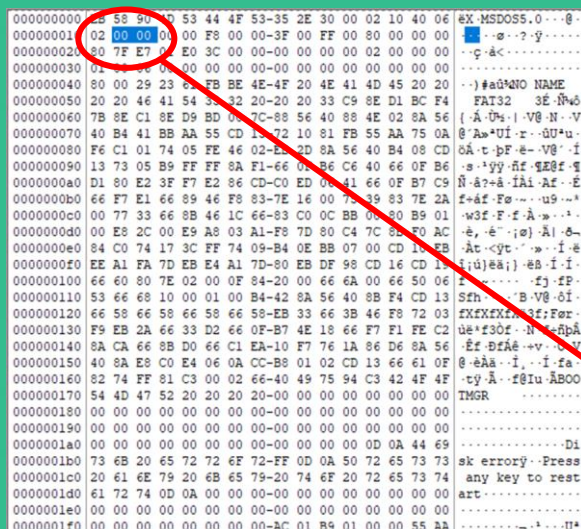

$$= 1024 + 512 + 64 = 1600$$

Size in sectors of the reserved area.

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. At the top, the letters 'NFSU' are written in a serif font. The shield is divided into four quadrants: the top-left quadrant contains a white computer monitor icon on a blue background; the top-right quadrant contains a white fingerprint icon on a blue background; the bottom-left quadrant contains a white icon of a DNA double helix on a blue background; and the bottom-right quadrant contains a white icon of a microscope on a blue background. The entire shield is set against a light blue background.

$$= 2$$


The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. At the top, the letters 'NFSU' are written in a bold, sans-serif font. The shield is divided into four quadrants: the top-left quadrant contains a white computer monitor icon on a dark blue background; the top-right quadrant contains a white fingerprint icon on a dark blue background; the bottom-left quadrant contains a white icon of a microscope on a dark blue background; and the bottom-right quadrant contains a white DNA double helix icon on a dark blue background. The entire shield is set against a light blue background.



This is 0 for FAT₃₂ and typically 512 for FAT₁₆.

FAT: Structure Analysis (19-20)

```
00000000 EB 58 00 4D 53 4F 53-35 2E 30 00 02 10 40 06 eX-MSDOS5.0...
00000001 02 00 00 00 F8 00 00-3F 00 FF 00 80 00 00 00 ..ø-?y-...
00000002 80 7F 01 E0 3C 00 00-00 00 00 00 02 00 00 00 ..ç-à<...
00000003 01 00 06 00 00 00 00-00 00 00 00 00 00 00 00 .....
00000004 80 00 29 23 61 FB BE 4F 20 4E 41 4D 45 20 20 ..)##ANO NAME
00000005 20 20 46 41 54 33 32 20 20 33 C9 8E D1 BC F4 FAT32 3E 846
00000006 7B 8E C1 8E D9 BD 00 7C 80 56 40 88 4E 02 8A 56 {ÀÜ-|V8-N-V
00000007 40 B4 41 BB AA 55 CD 13-72 10 81 FB 55 AA 75 0A 0'As*Uf-r-GU*U
00000008 F6 C1 01 74 05 FE 46 02-EB 2D 80 56 40 B4 08 CD 5A-t:pF-e-V8'i
00000009 13 73 05 B9 FF FF 8A F1-66 0F B6 C0 40 66 0F B6 -s-yg-Nf-SEBf-g
0000000a D1 80 E2 3F F7 E2 86 CD-C0 ED 06 41 80 0F B7 C9 N-À7-à-IAi-Af-E
0000000b 66 F7 E1 66 89 46 F8 83-7E 16 00 75 39 7E 2A f-áf-Fo--u9--*
0000000c 00 77 33 66 8B 46 1C 66-83 C0 0C BB 00 80 20 01 -w3f-F-f-À-...
0000000d 00 E8 2C 00 E9 A8 03 A1-F8 7D 80 C4 7C 8B F0 7A -e,-é-[s]Ài-è-
0000000e 84 C0 74 17 3C FF 74 09-B4 0E BB 07 00 CD 10 EB -At-<yt--i-e
0000000f EE A1 FA 7D EB E4 A1 7D-80 EB DF 98 CD 16 CD 19 iju-88j]-eA-i-f-
00000010 66 60 80 7E 02 00 0F 84-20 00 66 6A 00 66 50 06 f--...fj-fP-
00000011 53 66 68 10 00 01 00 B4-42 8A 56 40 8B F4 CD 13 Sth-----U8-6i-
00000012 66 58 66 58 66 58 66 58-EB 33 66 3B 46 F8 72 03 fxfxfxfxfxf-8er-
00000013 F9 EB 2A 66 33 D2 66 0F-B7 4E 18 66 F7 F1 FE C2 0a*930f-N-f-m-8
00000014 8A CA 66 8B D0 66 C1 EA-10 F7 76 1A 86 D6 8A 56 Èf-BfAe-vv-O-V
00000015 40 8A E8 C0 E4 06 0A CC-B8 01 02 CD 13 66 61 0F 0-eAa-i,-i-fa-
00000016 22 74 FF 81 C3 00 02 66-40 49 75 94 C3 42 4F 4F -ty-Å-f8Iu-ÀBOO
00000017 54 4D 47 52 20 20 20 20-00 00 00 00 00 00 00 00 TMGR .....
00000018 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
00000019 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000001a 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000001b 73 6B 20 65 72 72 6F 72-FF 0D 0A 50 72 65 73 73 sk errorry--Press
0000001c 20 61 6E 79 20 6B 65 79-20 74 6F 20 72 65 73 74 any key to rest
0000001d 61 72 74 0D 0A 00 00 00-00 00 00 00 00 00 00 00 art .....
0000001e 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000001f 00 00 00 00 00 00 00 00-AC 01 B9 01 00 00 55 AA .....U*
```

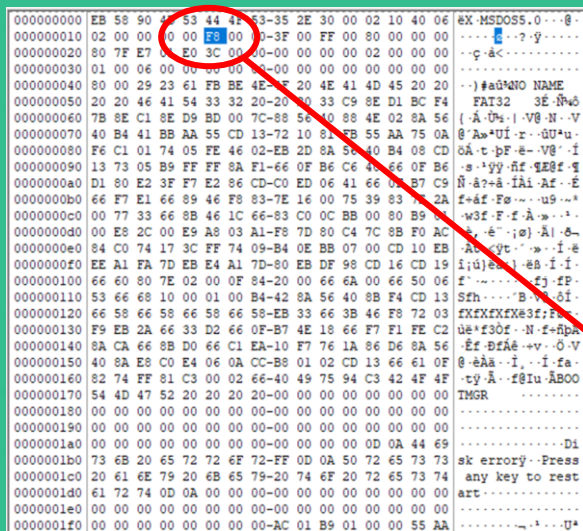


```
00 00 F8 00 00-3F 00 FF 00 80 00 00 00 00 ..ø-?y-...
01 E0 3C 00 00-00 00 00 00 00 02 00 00 00 ..ç-à<...
00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
```

16-bit value of the number of sectors in a file system.

If the number of sectors is larger than can be represented in this 2-byte value, a 4-byte value exists later in the data structure, and this should be 0.

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central blue field divided into four quadrants. The top-left quadrant contains a white icon of a computer monitor with a magnifying glass. The top-right quadrant contains a white fingerprint icon. The bottom-left quadrant contains a white icon of a circular arrow, symbolizing a process or cycle. The bottom-right quadrant contains a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in a bold, black, sans-serif font. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a black, Devanagari script.



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The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

[illegible]

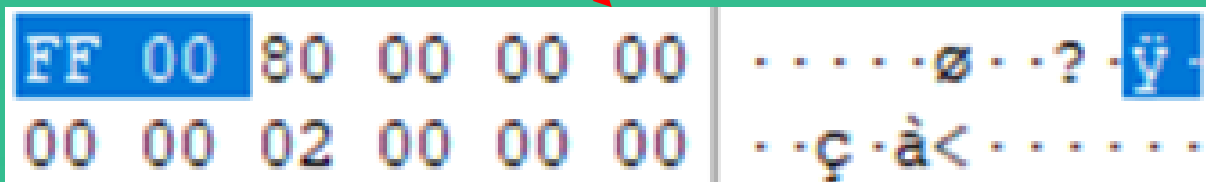
00	00-3F	00	FF	00	80	00	00	00	e-?
00	00-00	00	00	00	02	00	00	00	..ç-à<...
00	00-00	00	00	00	00	00	00	00	

16-bit size in sectors of each FAT for FAT₁₂ and FAT₁₆.
For FAT₃₂, this field is 0.

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$$= 32 + 16 + 8 + 4 + 2 + 1 = 63$$


The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

$$= 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$
$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$


National Forensic Sciences University

The logo of the National Food Security University (NFSU) is a shield-shaped emblem. At the top, the acronym 'NFSU' is written in a serif font. The shield is divided into four quadrants: top-left (blue) with a white computer monitor icon, top-right (red) with a white fingerprint icon, bottom-left (red) with a white circular arrow icon, and bottom-right (blue) with a white DNA double helix icon. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a stylized Devanagari script.

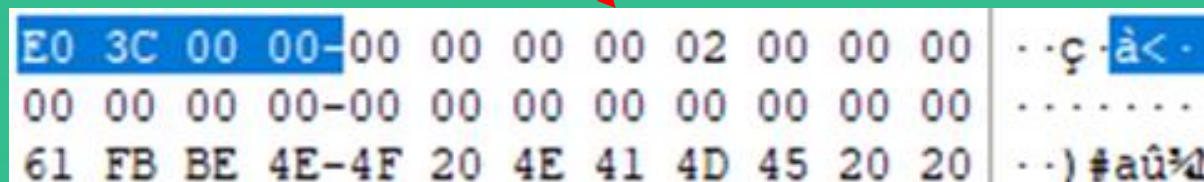
$$= 1 * 2^7 + 1 * 2^6 + 1 * 2^5 + 1 * 2^4 + 1 * 2^3 + 1 * 2^2 + 1 * 2^1 + 1 * 2^0$$
$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$


The logo of NFSU (National Food Security University) is a shield-shaped emblem. It features a yellow shield with a blue border. Inside the shield, there are four quadrants: top-left (blue with a white computer monitor icon), top-right (blue with a white fingerprint icon), bottom-left (red with a white circular arrow icon), and bottom-right (red with a white DNA double helix icon). Above the shield, the letters 'NFSU' are written in blue. Below the shield, the motto 'विद्यया अमृतं अश्नुते' is written in Devanagari script in blue.



The logo of NFSU (National Food Security University) is a shield-shaped emblem. It features a yellow shield with a blue border. Inside the shield, there are four quadrants: top-left (blue with a white laptop icon), top-right (red with a white fingerprint icon), bottom-left (blue with a white circular arrow icon), and bottom-right (red with a white DNA double helix icon). Above the shield, the letters 'NFSU' are written in blue. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in blue Devanagari script.

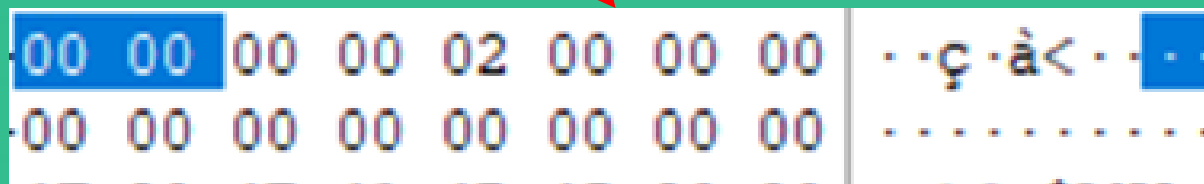
Output: x00 x00 x3C xE0 = 0000 0000 0011 1100 1110 0000
= 15,584



National Forensic Sciences University

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

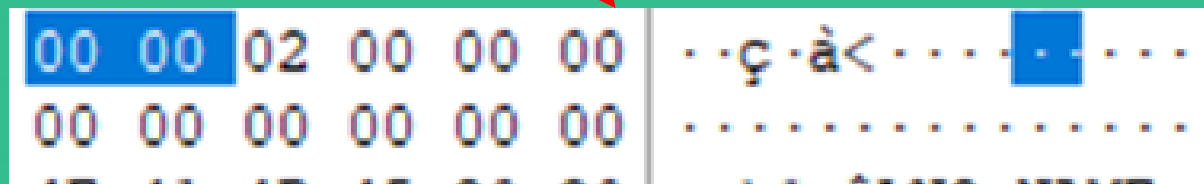
How to Calculate: Value is 00 00



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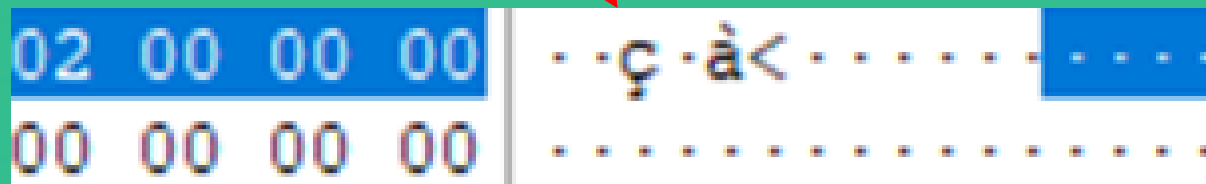
How to Calculate: Value is 00 00



National Forensic Sciences University

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidya Ya Amritam Ashnute) is written in yellow.

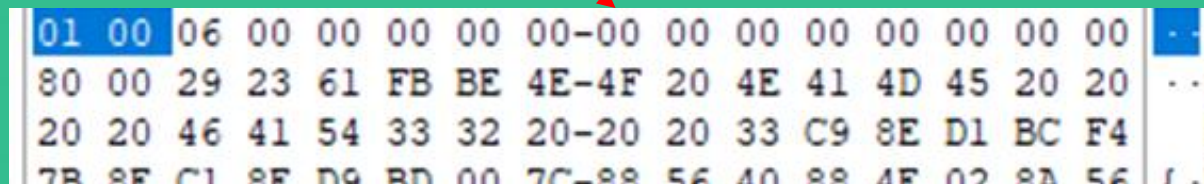
Output: x00 x00 x00 x02 = 0000 0000 0000 0010
= 1 * 2^1 = 2



National Forensic Sciences University

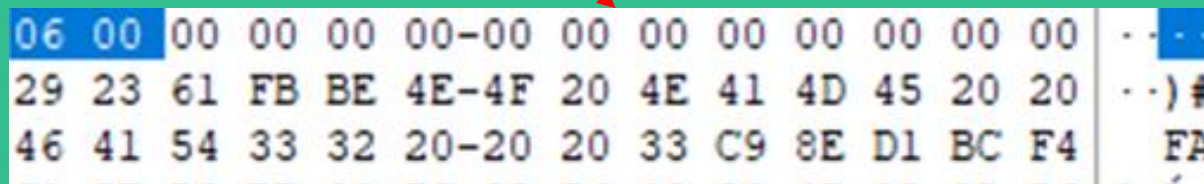
The logo of the National Food Security University (NFSU) is a shield-shaped emblem. At the top, the acronym 'NFSU' is written in blue. The shield is divided into four quadrants: top-left (blue with a white computer monitor icon), top-right (red with a white fingerprint icon), bottom-left (red with a white circular arrow icon), and bottom-right (blue with a white DNA double helix icon). Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyāya amṛtaṁ aśnute) is written in a blue, curved banner.

Output: $x_0 x_1 = 0000\ 0000\ 0000\ 0001$
 $= 1 * 2^0 = 1$



National Forensic Sciences University

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

$$= 1 * 2^2 + 1 * 2^1 = 4 + 2 = 6$$


National Forensic Sciences University

FAT32: Structure Analysis (52-63)

```
00000000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06 EX-MSDOSS.0-...0-
0000000010 02 00 00 00 00 00 F8 00 00-3F 00 FF 00 80 00 00 00 .....s-?y-...
0000000020 80 7F E7 01 50 00 00-00 00 00 00 00 00 00 00 .....g-ák-.....
0000000030 01 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
0000000040 80 00 29 00 00 00 00 00-4F 20 4E 41 4D 45 20 20 .....) #aû%NO NAME
0000000050 20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4 FAT32 3É-Ñ4ô
0000000060 7B 8E C1 8E D9 BD 00 7C-88 56 40 88 4E 02 8A 56 (Á Ú4-| V8-N-V
0000000070 40 B4 41 BB AA 55 CD 13-72 10 81 FB 55 AA 75 0A Áw*Uí r -0U*u-
0000000080 F6 C1 01 74 05 FE 46 02-EB 2D 8A 56 40 B4 08 CD 00 t.pF-e- V8'i
0000000090 13 73 05 B9 FF FF 8A F1-66 0F B6 C6 40 66 0F B6 -s-00-ff-1E8f-í
00000000a0 D1 80 E2 3F F7 E2 86 CD-C0 ED 06 41 66 0F B7 C9 N-á-íÁi Af-É
00000000b0 66 F7 E1 66 89 46 F8 83-7E 16 00 75 39 83 7E 2A f-áf-F-úS-...
00000000c0 00 77 33 66 8B 46 1C 66-83 C0 0C BB 00 80 B9 01 w3f-F-Á-...
00000000d0 00 E8 2C 00 E9 A8 03 A1-F8 7D 80 C4 7C 8B F0 AC -e,-é-í-0-Ái-0-
00000000e0 84 C0 74 17 3C FF 74 09-B4 0E BB 07 00 CD 10 EB Át-<9t-...í-é
00000000f0 EE A1 FA 7D EB E4 A1 7D-80 EB DF 98 CD 16 CD 19 í;ú|eá| -é-í-í-
0000000100 66 60 80 7E 02 00 0F 84-20 00 66 6A 00 66 50 06 f'-.....f3-0F-
0000000110 53 66 68 10 00 01 00 B4-42 8A 56 40 8B F4 CD 13 Sth-...B-V8-0-
0000000120 66 58 66 58 66 58 66 58-EB 33 66 3B 46 F8 72 03 fXfXfXfXfXf;Fer-
0000000130 F9 EB 2A 66 33 D2 66 0F-B7 4E 18 66 F7 F1 FE C2 úe*f30f-N-f-0pÁ
0000000140 8A CA 66 8B D0 66 C1 EA-10 F7 76 1A 86 D6 8A 56 -Éf-0fÁ-+v-0-V
0000000150 40 8A E8 C0 E4 06 0A CC-B8 01 02 CD 13 66 61 0F 0 -éÁ-í-í-f-á-
0000000160 82 74 FF 81 C3 00 02 66-40 49 75 94 C3 42 4F 4F -ty-Á-f0íu-ÁBOO
0000000170 54 4D 47 52 20 20 20 20-00 00 00 00 00 00 00 00 TMGR
0000000180 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000000190 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
00000001a0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
00000001b0 73 6B 20 65 72 72 6F 72-FF 0D 0A 50 72 65 73 73 sk errorÿ-Press
00000001c0 20 61 6E 79 20 6B 65 79-20 74 6F 20 72 65 73 74 any key to rest
00000001d0 61 72 74 0D 0A 00 00 00-00 00 00 00 00 00 00 00 art-.....
00000001e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
00000001f0 00 00 00 00 00 00 00 00-AC 01 B9 01 00 00 55 AA .....-...U*
```

```
00 00 00 00-00 00 00 00 00 00 00 00 00 00 .....
61 FB BE 4E-4F 20 4E 41 4D 45 20 20 ..) #aû%NO NAME
54 33 32 20-20 20 33 C9 8E D1 BC F4 FAT32 3É-Ñ4ô
D9 BD 00 7C-88 56 40 88 4E 02 8A 56 (Á Ú4-| V8-N-V
```

Reserved.



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How to Calculate:
Value is 80
Output: $x_{80} = 1000\ 0000$
 $= 1 * 2^7 = 128$



80	00	29	23	61	FB	BE	4E-4F	20	4E	41	4D	45	20	20	
20	20	46	41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4	
7B	8E	C1	8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56	{
46	24	41	2E	33	33	33	33-33	33	33	33	33	33	33	33	

BIOS INT13h drive number.

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.



```

000000000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06 eX MSDOS5.0 - 8
000000010 02 00 00 00 00 00 F8 00 00-00 3F 00 FF 00 00 00 00 00 00 00 00 00
000000020 80 7F E7 01 E0 3C 00 00-00 00 00 00 02 00 00 00 00 00 00 00 00 00
000000030 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00
000000040 00 20 1A 41 54 33 20 20-20 33 C9 8E D1 BC FA
000000050 7B 5E C1 E9 DB D0 7C-58 56 48 08 02 8A 5E
000000060 04 B4 41 8A 55 CD 13-72 10 81 F5 8A 75 0A
000000070 F6 C1 01 74 09 F6 4E-02 EB 2D 8A 56 04 B4 08 CD
000000080 13 73 05 B9 FF 8A F1-66 0F B6 C 6 06 0F B6
000000090 80 E2 3F 82 28 3D CD-ED 06 75 6F 0F B6 C9
0000000A0 EF E2 3F 82 28 3D-76 16 75 6F 0F B6 C9
0000000B0 00 77 33 6B 4E 1C-6E 0C BB 00 80 B5 01
0000000C0 0E 2C 00 E9 83 03-81 F8 04 7C 78 2A 8C
0000000D0 34 C0 74 33 FF 74-09 B4 08 07 00 02 10 EB
0000000E0 EE A1 FA 7D EB EA 1-7D 80 EB DF CD 16 CD 1B
0000000F0 66 60 80 7E 02 0F-84 20 00 66 80 66 50 0E
000000100 53 6E 10 00 00 01 00-8A 42 8A 56 04 B5 0A CD
000000110 68 58 68 58 68 58-68 58 33 68 58 33 68 58
000000120 FE 2A 66 33 D2 6E-0F B7 4E 18 66 F7 F1 F6
000000130 8A 68 6B D0 6B C1-EA 10 F7 76 1A 86 D6 8A 5E
000000140 0A EA E8 0C E4 06-CC B8 01 CD 13 66 01 F1
000000150 82 74 FF 81 C3 00-02 66 40 49 75 94 C3 42 4F 4F
000000160 54 4D 47 52 20 20-20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
000000170 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
000000180 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
000000190 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000001A0 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000001B0 73 EB 20 65 72 72-6F 72 FF 0D 0A 50 72 65 73 73
0000001C0 20 61 6E 79 0D 6B-65 79 20 74 6F 20 72 65 73 74
0000001D0 61 72 74 7D 0A 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000001E0 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000001F0 00 00 00 00 00 00-00 00 AC 01 B9 01 00 55 AA

```

00	29	23	61	FB	BE	4E-4F	20	4E	41	4D	45	20	20
20	46	41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4
8E	C1	8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56

Not used.

FAT32: Structure Analysis (65-65)

```
00000000 EB 59 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06 6X MSDOS5.0...8-
00000001 02 00 00 00 00 00 F8 00 00-3F 00 FF 00 80 00 00 00 ...s-?y...
00000002 80 7F E7 01 E0 3C 00 00-00 00 00 00 02 00 00 00 ...g-á...
00000003 01 7F E0 00 00 00 00 00-00 00 00 00 00 00 00 00 ...
00000004 00 00 29 23 61 FB BE 4E-4F 20 4E 41 4D 45 20 20 ...fa0MNO NAME
00000005 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4 FAT32 3E 8w6
00000006 7B 3E C1 8E D9 BD 00 7C-88 56 40 88 4E 02 8A 56 { .Á
00000007 40 B4 41 BB AA 65 CD 13-72 10 81 FB 55 AA 75 0A 8 Aw*Uf r-8U*U
00000008 F6 C1 01 74 05 11 46 02-EB 2D 8A 56 40 B4 08 CD 8A t.jpF-e-V8*V
00000009 13 73 05 B9 FF FF 8E F1-66 0F B6 C6 40 66 0F B6 s-yp-af-8E8f-
0000000a D1 80 E2 3F F7 E2 86 C0 ED 06 41 66 0F B7 C9 N Á?~ã IÁI Af-É
0000000b 66 F7 E1 66 89 46 F8 83-16 00 75 39 83 7E 2A f-af-Fe--us-~*
0000000c 00 77 33 66 8B 46 1C 66-83 0C BB 00 80 B9 01 w3f-F-f-Á-~*
0000000d 00 E8 2C 00 E9 A8 03 A1-F8 7D 00 C4 7C 8B FO AC e, é" [e] Á| e-
0000000e 84 C0 74 17 3C FF 74 09-B4 0E BB 00 00 CD 10 EB Át-<yt-~*~i f e
0000000f EE A1 FA 7D EB E4 A1 7D-80 EB DF 96 0D 16 CD 19 i;djæa;] e-a-i-f-
00000010 66 60 80 7E 02 00 0F 84-20 00 66 6A 00 66 50 06 f-~--~ f; fF-
00000011 53 66 68 10 00 01 00 B4-42 8A 56 40 8B F6 CD 13 Sth-~B-V8-6i-
00000012 66 59 66 58 66 58 66 58-EB 33 66 3B 46 F8 76 03 fXfXfXfXfXf3f;Fer-
00000013 F9 EB 2A 66 33 D2 66 0F-B7 4E 18 66 F7 F1 FE 0a*f30f-N-f-RpA
00000014 8A CA 66 8B D0 66 C1 EA-10 F7 76 1A 86 D6 8A 56 f-BrAe-+v-Ó V
00000015 40 8A E8 C0 E4 06 0A CC-B8 01 02 CD 13 66 61 0F 8 8A i, i fa-
00000016 82 74 FF 81 C3 00 02 66-40 49 75 94 C3 42 4F 4F ty-Á f8Iu-ÁBOO
00000017 54 4D 47 52 20 20 20 20-00 00 00 00 00 00 00 00 TMGR
00000018 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 ...
00000019 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 ...
0000001a 00 00 00 00 00 00 00 00-00 00 00 0D 0A 44 69 ...D1
0000001b 73 6B 20 65 72 72 6F 72-FF 0D 0A 50 72 65 73 73 sk errorÿ-Press
0000001c 20 61 6E 79 20 6B 65 79-20 74 6F 20 72 65 73 74 any key to rest
0000001d 61 72 74 0D 0A 00 00 00-00 00 00 00 00 00 00 00 art-~--~
0000001e 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 ...
0000001f 00 00 00 00 00 00 00 00-AC 01 B9 01 00 00 55 AA .....~*U*
```

```
29 23 61 FB BE 4E-4F 20 4E 41 4D 45 20 20 . .
46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4 FA
C1 8E D9 BD 00 7C-88 56 40 88 4E 02 8A 56 { .Á
```

Extended boot signature to identify if the next three values are valid.
The signature is 0x29.



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```

0000000000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 10 40 06 EX MSDOSD.0 - 8
0000000010 02 00 00 00 00 F8 00 00-3F 00 FF 00 00 00 00 .s - ? y...
0000000020 70 F7 E7 01 E0 3C 00 00-00 00 00 02 00 00 00 c a
0000000030 01 00 00 00 00 00 00 00-00 00 00 00 00 00 00
0000000040 80 00 00 00 00 00 00 00-00 00 00 00 00 00 00
0000000050 20 40 4E 5A 83 31 70-20 20 33 9E D1 BC FA [tab] NO NAME
0000000060 7B 8E C1 8E DB 0D 00 78-58 56 40 02 8A 5E 40 FAT32 3E -NVO
0000000070 40 B4 41 BB 8A 55 CD 13-10 1F 55 8A 75 0A 40 B UI r -DU+U
0000000080 06 F6 C1 01 74 4E 42-E8 10 4A 56 40 8B CD 0A t p k e -VS i
0000000090 13 05 E8 39 FF 42 86 F1-66 09 0E C6 46 0B 0B 0A t y f k -EER i
00000000A0 D1 80 E2 8F F7 42 86 F1-66 09 0E C6 46 0B 0B 0A N a n d -I a l a r E
00000000B0 66 F7 E1 80 42 86 F1-66 09 0E C6 46 0B 0B 0A f a f p e -u s
00000000C0 00 77 33 6B 4E 1C 66-93 00 CB 0B 09 B9 01 w3F F f A -s
00000000D0 08 E8 2C 00 E8 A8 03-41 F8 7D 80 4A 7C 8B F0 AC e f (s) A -e
00000000E0 84 C0 74 17 3C FF 74-09 B4 0E BB 07 00 00 00 00 AB C C y t -i f a
00000000F0 EE A1 7A 7D E8 E4 A1-7D 80 E8 F9 9C 16 80 19 i d (e a) e s i f -
0000000100 66 60 80 7E 02 00 01 00-84 20 66 6A 00 66 50 80 f f f p
0000000110 53 68 68 10 00 01 00 00-84 A2 8A 56 40 8B F4 CD 13 f h -B V8 0 i
0000000120 58 68 68 58 68 58 66-58 58 33 66 3B 46 F8 72 03 i n X X X X 3 f -R
0000000130 F9 EB 2A 66 33 D2 66-0F B7 4E 18 66 F7 F2 C2 u e f -N f -R p a
0000000140 8A CA 66 8B 0D 06 01-EA 10 F7 76 1A 82 66 8A 5E E B M -f -O V
0000000150 04 EA E8 00 EA 66 0A-CC B8 01 CD 13 66 01 f 0 e A i f f -f
0000000160 82 74 FF 81 C3 02 06-66 40 49 75 94 C3 42 4F 4F t y f -f @ I n B R O
0000000170 54 4D 47 52 20 20 20-00 00 00 00 00 00 00 00 I M G R
0000000180 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000190 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000001A0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000001B0 73 6B 20 65 72 72 6F-72 FF 0D 0A 50 72 65 73 73 s k e r r o y -P r e s s
00000001C0 20 61 6E 79 20 6B 65-79 20 74 6F 20 72 65 73 74 a n y k e y t o r e s t
00000001D0 61 72 74 0D 0A 00 00-00 00 00 00 00 00 00 00
00000001E0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000001F0 00 00 00 00 00 00-AC 01 B9 01 00 55 AA

```

23	61	FB	BE	4E-4F	20	4E	41	4D	45	20	20	..)	#aû
41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4		FAT32
8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56	{	·Á·Û

sarang.rajvansh@nfsu.ac.in

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central field divided into four quadrants. The top-left quadrant is blue with a white computer monitor icon. The top-right quadrant is red with a white fingerprint icon. The bottom-left quadrant is red with a white circular arrow icon. The bottom-right quadrant is blue with a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in yellow. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in yellow.

[illegible]

01 00 06 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
80 00 29 23 61 FB BE 4E-4F 20 4E 41 4D 45 20 20	..) #aũ%NO NAME
20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4	FAT32 3É-Ñ-ô
7B 8E C1 8E D9 BD 00 7C-88 56 40 88 4E 02 8A 56	{ .Á-Û- .V@-N- V
40 B4 41 BB BA 55 CD 13-72 10 81 FB 55 AA 75 0A	@'A-ũ-III- r- -ôII-u

The user chooses this value when creating the file system.

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central blue field divided into four quadrants. The top-left quadrant contains a white icon of a computer monitor with a magnifying glass. The top-right quadrant contains a white fingerprint icon. The bottom-left quadrant contains a white icon of a circular arrow, symbolizing a process or cycle. The bottom-right quadrant contains a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in a bold, black, sans-serif font. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a black, Devanagari script.

[illegible]

46	41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4	FAT32
C1	8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56	{ .Á.Ûs. .V
41	BB	AA	55	CD	13-72	10	81	FB	55	AA	75	0A	@'A»'uÍ.r.
01	74	05	EE	46	02-EB	2D	8A	56	40	B4	08	CD	8Á.r.bE.ä

Standard values include “FAT₃₂,” but nothing is required.

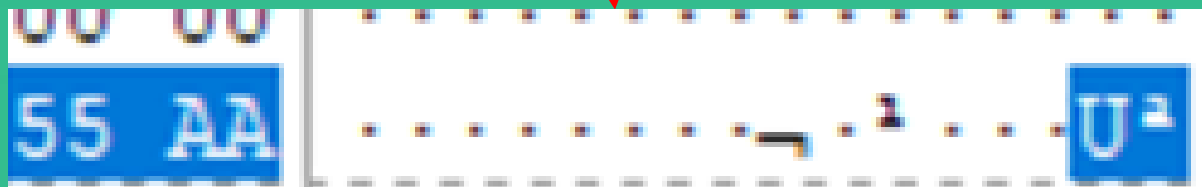
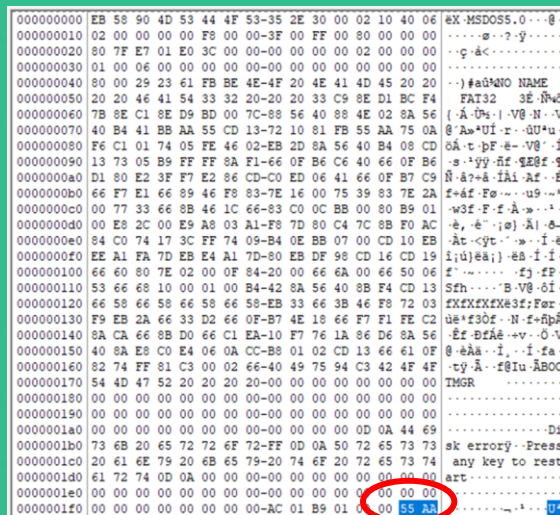
FAT32: Structure Analysis (90-509)

0000000000	EB	58	90	4D	53	44	4F	53-35	2E	30	00	02	10	40	06	eX-MSDOSS.0...@-
0000000010	02	00	00	00	00	F8	00	00-3F	00	FF	00	80	00	00	00	ø...?..ÿ.....
0000000020	80	7F	E7	01	E0	3C	00	00-00	00	00	00	02	00	00	00	..ç-à<.....
0000000030	01	00	06	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000000040	80	00	29	23	61	FB	BE	4E-4F	20	4E	41	4D	45	20	20	..)#aû%NO NAME
0000000050	20	20	46	41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4	FAT32 3E-N%ô
0000000060	7B	8E	C1	8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56	{-Ã-Û%- ..V@-N..V
0000000070	40	B4	41	BB	AA	55	CD	13-72	10	81	FB	55	AA	75	0A	@'A»²Uí-r..ûU²u.
0000000080	F6	C1	01	74	05	FE	46	02-EB	2D	8A	56	40	B4	08	CD	ôÃ-t-pF-è-..V@-í
0000000090	13	73	05	B9	FF	FF	8A	F1-66	0F	B6	C6	40	66	0F	B6	-s-²ÿÿ-ñf-¶E@f-¶
00000000a0	D1	80	E2	3F	F7	E2	86	CD-C0	ED	06	41	66	0F	B7	C9	N-â?+â-íÃí-Af-..É
00000000b0	66	F7	E1	66	89	46	F8	83-7E	16	00	75	39	83	7E	2A	f+áf-Fø-..u9-..²
00000000c0	00	77	33	66	8B	46	1C	66-83	C0	0C	BB	00	80	B9	01	..w3f-F-f-Ã-»-..²
00000000d0	00	E8	2C	00	E9	A8	03	A1-F8	7D	80	C4	7C	8B	F0	AC	..è, -é-..;ø} -Ãí-8-
00000000e0	84	C0	74	17	3C	FF	74	09-B4	0E	BB	07	00	CD	10	EB	..Àt-<ÿt-..-»-..í-è
00000000f0	EE	A1	FA	7D	EB	E4	A1	7D-80	EB	DF	98	CD	16	CD	19	í;ú}èä;}-ès-í-í-
0000000100	66	60	80	7E	02	00	0F	84-20	00	66	6A	00	66	50	06	f-..-..-fj-fP-
0000000110	53	66	68	10	00	01	00	B4-42	8A	56	40	8B	F4	CD	13	Sfh-..-..²B-V@-ôí-
0000000120	66	58	66	58	66	58	66	58-EB	33	66	3B	46	F8	72	03	fxfxfxfxe3f;Før-
0000000130	F9	EB	2A	66	33	D2	66	0F-B7	4E	18	66	F7	F1	FE	C2	ûè*f3òf-..N-f+ñpÃ
0000000140	8A	CA	66	8B	D0	66	C1	EA-10	F7	76	1A	86	D6	8A	56	..Èf-ðfÃè-+v-..Ö-V
0000000150	40	8A	E8	C0	E4	06	0A	CC-B8	01	02	CD	13	66	61	0F	@-èÃà-..í-..-í-fa-
0000000160	82	74	FF	81	C3	00	02	66-40	49	75	94	C3	42	4F	4F	..ty-Ã-..f@Iu-ÃBOO
0000000170	54	4D	47	52	20	20	20	20-00	00	00	00	00	00	00	00	TMGR
0000000180	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000000190	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
00000001a0	00	00	00	00	00	00	00	00-00	00	00	00	0D	0A	44	69	Di
00000001b0	73	6B	20	65	72	72	6F	72-FF	0D	0A	50	72	65	73	73	sk errorÿ-..Press
00000001c0	20	61	6E	79	20	6B	65	79-20	74	6F	20	72	65	73	74	any key to rest
00000001d0	61	72	74	0D	0A	00	00	00-00	00	00	00	00	00	00	00	art.....
00000001e0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
00000001f0	00	00	00	00	00	00	00	00-AC	01	B9	01	00	00	55	AA	U²

Not used.



The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. At the top, the letters 'NFSU' are written in a bold, sans-serif font. Below the text, the shield is divided into four quadrants. The top-left quadrant is blue and contains a white icon of a computer monitor with a magnifying glass over it. The top-right quadrant is red and contains a white fingerprint icon. The bottom-left quadrant is blue and contains a white icon of a microscope. The bottom-right quadrant is red and contains a white DNA double helix icon. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a stylized, curved font.



National Forensic Sciences University

The logo of NFSU (National Food Security University) is a shield-shaped emblem. It features a yellow shield with a blue border. Inside the shield, there are four quadrants: top-left (blue with a white computer monitor icon), top-right (red with a white fingerprint icon), bottom-left (red with a white circular arrow icon), and bottom-right (blue with a white DNA double helix icon). Above the shield, the text 'NFSU' is written in blue. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a blue, curved banner.



```
00000002 52 52 61 41 00 00 00 00-00 00 00 00 00 00 00 00 RBA
0000000210 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000220 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000230 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000240 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000250 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000260 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000270 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000280 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000290 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000002a0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000002b0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000002c0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000002d0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000002e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000002f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000300 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000310 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000320 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000330 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000340 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000350 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000360 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000370 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000380 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000000390 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000003a0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000003b0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000003c0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000003d0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
00000003e0 00 00 00 00 72 72 41 61-9C EF 1E 00 06 00 00 00
00000003f0 00 00 00 00 00 00 00 00-00 00 00 00 00 55 AA rrAa o
```

52	52	61	41	00	00	00	00-00	00	00	00	00	00	00	00	RRaP
00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	

Signature (0x41615252).

FAT32: Structure Analysis

FSINFO (4-483)

000000200	52	52	61	41	00	00	00	00-00	00	00	00	00	00	00	00	RRaA
000000210	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000220	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000230	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000240	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000250	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000260	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000270	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000280	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000290	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000002a0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000002b0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000002c0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000002d0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000002e0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000002f0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000300	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000310	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000320	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000330	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000340	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000350	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000360	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000370	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000380	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
000000390	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000003a0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000003b0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000003c0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000003d0	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00	
0000003e0	00	00	00	00	72	72	41	61-9C	6F	1E	00	06	00	00	00	rrAa -o -
0000003f0	00	00	00	00	00	00	00	00-00	00	00	00	00	55	AA		U²

Not used



FAT32: Structure Analysis

FSINFO (484-487)

```
000000200 52 52 61 41 00 00 00-00 00 00 00 00 00 00 00 RRaA .....
000000210 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000220 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000230 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000240 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000250 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000260 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000270 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000280 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000290 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000002a0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000002b0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000002c0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000002d0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000002e0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000002f0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000300 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000310 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000320 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000330 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000340 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000350 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000360 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000370 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000380 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
000000390 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000003a0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000003b0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000003c0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000003d0 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 .....
0000003e0 00 00 00 72 72 41 61-9C 6F 1E 00 06 00 00 00 rrAa .....
0000003f0 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA .....U*
```

```
00 00 00 00-00 00 00 00 00 00 00 00 00 .....
72 72 41 61-9C 6F 1E 00 06 00 00 00 00 ...rrAa
00 00 00 00-00 00 00 00 00 00 00 55 AA .....U*
```

Signature (ox61417272).



FAT32: Structure Analysis

FSINFO (488-491)

000000200	52 52 61 41 00 00 00-00 00 00 00 00 00 00	RRaA
000000210	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000220	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000230	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000240	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000250	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000260	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000270	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000280	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000290	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000002a0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000002b0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000002c0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000002d0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000002e0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000002f0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000300	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000310	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000320	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000330	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000340	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000350	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000360	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000370	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000380	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
000000390	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000003a0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000003b0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000003c0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000003d0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0000003e0	00 00 00 72 72 41 9C 6F 1E 00 06 00 00 00	rrAa:o
0000003f0	00 00 00 00 00 00 00-00 00 00 00 00 00 00	

00 00 00 00 00 00 00 00	
9C 6F 1E 00 06 00 00 00	rrAa:o

Number of free clusters



FAT32: Structure Analysis

FSINFO (492-495)

000000200	52 52 61 41 00 00 00-00 00 00 00 00 00 00 00 00 00 00	RRaA.....
000000210	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000220	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000230	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000240	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000250	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000260	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000270	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000280	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000290	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000002a0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000002b0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000002c0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000002d0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000002e0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000002f0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000300	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000310	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000320	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000330	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000340	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000350	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000360	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000370	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000380	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
000000390	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000003a0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000003b0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000003c0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000003d0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	
0000003e0	00 00 00 72 72 41 61-9C 6F 00 06 00 00 00 00 00 00 00	rrAa-o-.....
0000003f0	00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00	U*

06 00 00 00	rrAa-o-.....
00 00 55 AA	U*

Next free cluster



FAT32: Structure Analysis

FSINFO (496-507)

```
000000200 52 52 61 41 00 00 00-00 00 00 00 00 00 00 00 00 RRaA .....
000000210 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
000000220 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
000000230 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
000000240 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
000000250 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....
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Not used.



FAT32: Structure Analysis

FSINFO (508-511)

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06 00 00 00rrAa o
00 00 55 AAU

Signature (0xAA550000).



TASK



File Allocation Table Analysis



References



- File System Forensics by Brian Carrier

